
Scaling Empowered Agents

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Abstract

Empowerment measures the capacity of an agent to actively control its environment. This object has been identified as an unsupervised mechanism for natural and artificial agents to make sense of large environments. In this work, we scale empowerment maximization to large discrete environments and high-dimensional, continuous environments. In addition to exhibiting exploration behavior, empowered agents seek out states in the environment that grant high degrees of control. Our results also show that empowerment depends strongly on the underlying state representation and should not be arbitrarily maximized over full state spaces. Our work supports empowerment maximization as a principled mechanism to train unsupervised agents in large environments.

1 Introduction

In large state spaces, searching for sparse rewards is a hard, often impossible task. An undirected agent often cannot stumble upon interesting behavior using random actions. However, real environments have inherent structure that makes the problem of searching for sparse reward more tractable. For instance, an agent may wish to accumulate resources and gain control over certain objects in an environment, so these objects may serve as a resource for arbitrary future task; most interesting tasks in an environment (e.g. via externally-defined rewards) often align with the same degrees of freedom as resource accumulation. In this work, we introduce a method to train agents that maximizes the generic, universal resource of *control*: we reward an agent for its ability to exert control over an environment. We quantify control using the information-theoretic quantity of **empowerment**, and build on state-of-the-art skill learning methods to maximize empowerment in large environments.

Empowerment measures the degree to which an agent can control their environment. Prior literature has shown that empowerment maximization can help explain how natural and artificial agents build causal models of the world and interact with large environments [1–3]. Indeed, empowerment speaks to a general need for RL methods that are compatible with the Big World Hypothesis [4]: real-world agents should successfully interact with an environment that they cannot fully describe. Despite a long line of literature studying the behavior of empowerment-maximizing agents, there has not yet been empirical demonstrations of empowerment maximization in such challenging, large settings that escape simple characterization. Early literature (VIC [5] and its predecessor) estimate and maximize empowerment

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as a channel capacity using variational methods [6–8]. However, most of these works only scale up to simple tabular settings or low dimensional continuous settings. These methods eventually gave rise to a later line of unsupervised skill discovery literature successful in higher-dimensional locomotion and manipulation tasks [9? –12].

We aim to close this gap between concept and current practice. Our starting observation is that state-of-the-art mutual information skill learning methods already maximize quantities similar to empowerment; MISL maximizes value functions that lower bound empowerment. In this work, we present a recipe to adapt unsupervised skill learning methods to maximize empowerment in challenging, high-dimensional settings. The goal of this work is to provide methods to tractably train empowerment-maximizing agents. Using modern skill learning methods, we present strong exploration results in Craftax-Classic [13]. Furthermore, our empirical results show that empowerment maximization rewards behavior distinct from canonical intrinsic exploratory methods, prioritizing resource and skill accumulation over novelty.

Our work suggests that empowerment maximization is a tractable mechanism for agents to explore large environments. We present future directions to make empowerment maximization methods more robust and scalable, particularly in the continuous setting.

2 Related Work

Intrinsically-Motivated Exploration Methods. Intrinsically motivated methods are exploration methods in reinforcement learning (RL) that require no external information about the Markov decision process (MDP), outside of environment interaction. Such methods are classically used in settings with sparse or absent rewards, where intrinsic signals can help synthesize complex behaviors. Roughly categorized into curiosity, novelty, information gain, and prediction error. Typical methods optimize these intrinsic rewards alongside extrinsic rewards. Scalable methods include RND (prediction error to output of a randomly-initialized neural network), E3B (ellipticity metric that rewards exploration in low-variance directions), BYOL-Explore, temporal distances (e.g. Go-Explore (hard-to-reach goals), ETD, C-TeC). An entire suite of proposed exploration methods have strong theoretical properties but are challenging to optimize in high-dimensional settings—precisely the open-ended settings that would benefit from exploration beyond extrinsic rewards. These include methods based on novelty over state occupancy and reward maximization with local linearization arguments. Empowerment is one such object. Despite the promise of these methods, including standard objectives like RND, often show no benefits in benchmarks we may care about. There are immediately some problems – these objectives are often engineered to decay under repeated exposure to the same observations, and are by-definition non-stationary which, when combined with bootstrapping and function approximation, becomes problematic (e.g. some of these issues addressed in RND paper – for instance, separate value functions for intrinsic versus extrinsic rewards). In this work, we do not purport to solve these problems that are generically true of intrinsic rewards, and focus on the problem setting optimizing *the intrinsic empowerment signal independent of extrinsic signal*. Not a new problem setting (see CTeC paper, skill discovery paper, etc.).

Unsupervised Skill Discovery. Unsupervised skill discovery methods learn latent-conditioned policies without access to extrinsic (supervised) reward signals. Mutual Information Skill Learning (mutual information skill learning (MISL)) forms a subset of these methods, learning policies that maximize mutual information mutual information (MI) between observations and actions [9]. Generally, MISL methods learn policies such that learned latents are informative of future states.

Potential and Effective Empowerment. Classically, empowerment quantifies the influence an agent’s actions can exert on the future state distribution of the environment [7?]. This is formalized as a channel capacity (maximal directed information over actions) between the sequence of actions and future states [14]. As originally defined, potential empowerment is the channel capacity at a particular state, quantifying how much an agent can control from that state.

In works from psychology and cognitive sciences, empowerment has been conjectured to explain human behavior in complex, open-ended settings not captured by other intrinsic motivations like

curiosity [1, 3, 15, 16]. For example, an agent seeking to control its environment would not waste effort staring at random noise (the Noisy TV problem), a desirable property for both human-like and assistance agents [17, 18]. Several works have proposed empowerment and related objectives as suitable for training artificial agents.

Prior work on estimating and maximizing empowerment includes variational approaches like VIC [5] and related baselines [19], though these methods do not scale to high-dimensional continuous settings. Related unsupervised skill discovery methods include CSF/METRA, DIAYN, and GenDa.

Skill Learning and Effective Empowerment. MISL objectives have a close relationship to empowerment. Importantly, the effective empowerment objective and skill learning objectives are similar, even though modern skill learning methods do not explicitly reference empowerment. Indeed, predecessors to modern MISL methods—including Mohamed, Rezende, and VIC [5, 6]—explicitly aim to maximize forms of empowerment. Modern MISL methods optimize effective empowerment by learning skills that actively transmit information to the future.

Effective empowerment lower bounds potential empowerment [19]. However, prior empowerment maximization approaches did not scale to high-dimensional continuous settings. In this work, we aim to close this gap by optimizing empowerment using modern MISL methods in large, open-ended settings—the setting where empowerment matters. We introduce methods and provide both on-policy and off-policy implementations, highlighting key practical considerations: (1) empowerment over observation versus state space, (2) defining empowerment in continuous action spaces, and (3) breaking symmetry in empowerment-driven policies.

The key contribution of our paper is adapting modern MISL approaches to empowerment maximization, demonstrating that empowerment maximization is tractable in large, open-ended settings. Our work opens empowerment maximization as a practical objective and advances research on intrinsically-motivated agents in open-ended environments.

3 Preliminaries

MDP problem setting. We present notation for a fully observed MDP with states $s \in \mathcal{S}$, actions $a \in \mathcal{A}$, and environment dynamics $p(s' | s, a)$.

The latent skill z lies in a continuous skill space $\mathcal{Z} \subseteq \mathbb{R}^d$, taken to be the unit hypersphere in our continuous-control experiments. We fix $z \sim p(z) = \text{Unif}(\mathcal{Z})$ and write $\Pi_{\mathcal{Z}}$ for the class of skill- and start-state-conditioned policies $\pi(a | s, z; s_0)$, where s_0 is the context of the skill [20]. We write Π for policies $\pi(a | s)$ and fix a discount factor $\gamma \in (0, 1)$.

Shifted discounted state occupancy. For an initial state s_0 , skill $z \in \mathcal{Z}$, and policy $\pi(a | s, z; s_0)$, define the discounted state occupancy measure (DSOM)

$$p^\pi(s_+ = s | s_0, z) \triangleq (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t p^\pi(s_{t+1} = s | s_0, z). \quad (1)$$

Averaging over a fixed skill source gives

$$p^\pi(s_+ | s_0) = \mathbb{E}_{p(z)} [p^\pi(s_+ | s_0, z)]. \quad (2)$$

Potential and effective empowerment. We distinguish potential empowerment, the maximum achievable MI from a state, from effective empowerment, the MI realized by a fixed skill-conditioned policy [14, 21].

Definition 3.1 (Potential empowerment). Let $Z \sim p(z) = \text{Unif}(\mathcal{Z})$ and let $S_+ \sim p^\pi(s_+ | s_0, Z)$. Potential empowerment is

$$\mathcal{E}_{\text{pot}}(s_0) \triangleq \max_{\pi \in \Pi_{\mathcal{Z}}} I^\pi(Z; S_+ | S_0 = s_0). \quad (3)$$

For continuous $\mathcal{Z}$, fixing the uniform prior loses no generality. However, our experiments use finite set of skills for simplicity. This is a practical assumption that does lose generality, instead approximating channel capacity on a space of discretized source distributions.

Potential-policy value functions. For any policy $\pi \in \Pi$, define

$$Q_{\text{pot}}^{\pi}(s, a) \triangleq \mathbb{E}_{\pi} \left[\sum_{k=0}^{\infty} \gamma^k \mathcal{E}_{\text{pot}}(s_{t+k}) \mid s_t = s, a_t = a \right], \quad (4)$$

$$V_{\text{pot}}^{\pi}(s) \triangleq \mathbb{E}_{a \sim \pi(a|s)} [Q_{\text{pot}}^{\pi}(s, a)]. \quad (5)$$

The definition of a potential empowerment maximizing policy follows:

Definition 3.2 (Potential empowerment policy). A potential empowerment policy $\pi_{\text{pot}} \in \Pi$ takes the potential empowerment of Definition 3.1 as a reward.

$$\pi_{\text{pot}} \in \arg \max_{\pi \in \Pi} V_{\text{pot}}^{\pi}(s) \quad \text{for all } s \in \mathcal{S}. \quad (6)$$

While conceptually similar, mutual information skill learning methods do not maximize potential empowerment, and maximize effective empowerment. *Effective empowerment* captures the MI itself:

Definition 3.3 (Effective empowerment). For a fixed policy $\pi \in \Pi_{\mathcal{Z}}$, effective empowerment at s_0 is the MI realized under the same uniform skill source:

$$\mathcal{E}_{\text{eff}}(\pi, s_0) \triangleq I^{\pi}(Z; S_+ \mid S_0 = s_0) \quad (7)$$

$$= \mathbb{E}_{p(z)} D_{\text{KL}} [p^{\pi}(s_+ \mid s_0, z) \parallel p^{\pi}(s_+ \mid s_0)]. \quad (8)$$

Effective empowerment contains no maximization and measures the MI achieved by a particular policy. Consequently,

$$\mathcal{E}_{\text{pot}}(s_0) = \max_{\pi \in \Pi_{\mathcal{Z}}} \mathcal{E}_{\text{eff}}(\pi, s_0). \quad (9)$$

4 Methods

MISL methods learn skill-conditioned policies π_{eff} and value function that measure the *effective empowerment* from a state s_0 (Definition 3.3). These skills therefore serve as the yardsticks of empowerment: the diversity of these skills provides a lower bound on the potential empowerment of s_0 . A *potential empowerment* policy π_{pot} instead visits states with high potential empowerment:

$$\pi_{\text{pot}} \in \arg \max_{\pi \in \Pi} \mathbb{E}_{\pi} \left[\sum_t \gamma^t \mathcal{E}_{\text{pot}}(s_t) \right]. \quad (10)$$

This connection suggests a two-policy algorithm. The skill policy π_{eff} estimates a lower bound $\widehat{\mathcal{E}}_{\text{pot}}(s)$ on statewise potential empowerment, while the potential policy π_{pot} maximizes this estimate as an intrinsic reward. The effective and potential empowerment policies therefore support one another: the skills π_{eff} measure the controllable futures available from a state, while the potential empowerment policy π_{pot} moves the agent toward states from which the agent can exert control.

High-level method summary. The empowerment-maximization algorithm (Algorithm 1) alternates between training the two policies shown in Fig. 1. First,

Algorithm 1: **Empowerment maximization.** Skills estimate the degree of control from each starting state. The potential policy π_{pot} ascends this estimate. Full version in Algorithm 2.

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1: for  $n = 1, \dots, N$  do
2:   Skills: train  $\pi_{\text{eff}}$  using  $Q_{\text{eff},z}^{\pi_{\text{eff}}}$ 
3:   Estimate:  $\widehat{\mathcal{E}}_{\text{pot}}(s) \leftarrow \mathbb{E}_{p(z)} [V_{\text{eff},z}^{\pi_{\text{eff}}}(s; s)]$ 
4:   Ascend: train  $\pi_{\text{pot}}$  using  $\widehat{Q}_{\text{pot}}^{\pi_{\text{pot}}}$ 
    
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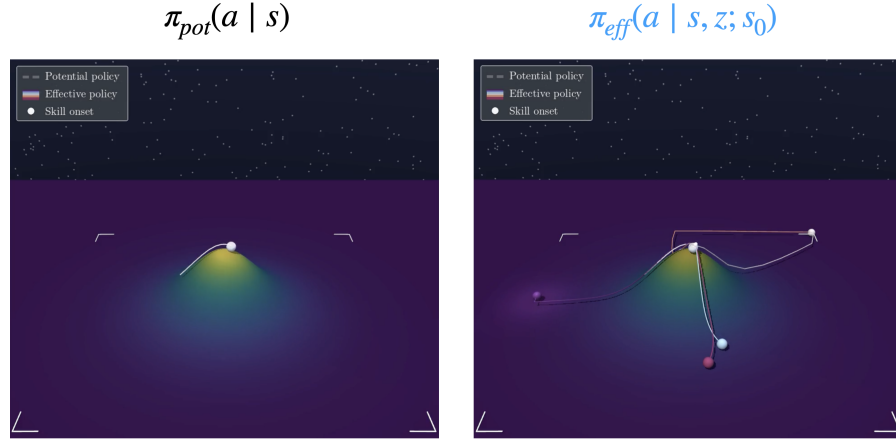


Figure 1: **Two trained empowerment-maximizing policies characterize central states in an MDP.** Consider the MDP of an agent (ball) on a hill, where the top represents a clearing. (Left) The potential empowerment policy (white) ascends the empowerment landscape, using the per-state empowerment $\mathcal{E}_{\text{pot}}(s)$ as an intrinsic reward. (Right) The effective empowerment policy π_{eff} (colors) maximizes $I^\pi(Z; S_+ | S_0)$ over skills z , and provides a lower bound on the potential empowerment.

we train the skills π_{eff} and use their on-policy values to estimate a lower bound $\widehat{\mathcal{E}}_{\text{pot}}(s)$ on potential empowerment. We then use this estimate as an intrinsic reward for the potential policy π_{pot} . For our chosen MISL objective, the skill update is an effective-empowerment policy-improvement step under the assumptions of Theorem A.1, a modified statement of the policy improvement results in [5]. Thus, MISL methods can lead to policy improvement on a lower bound of the potential empowerment.

Training structure. However, while we have a general idea of the objects to optimize, key questions remain in building a scalable implementation of empowerment maximization:

1. How do we gather data to train the empowerment objective?
2. What are practical implementation details necessary to get empowerment maximization methods to work in high DoF settings?

Switching policies. While we presented a methodology to convert skill-learning methods into potential empowerment maximization algorithms, we cannot simply learn skills for every starting state s_0 , bound the potential empowerment using the effective empowerment, then maximize the potential empowerment. For arbitrarily large state spaces, learning skills everywhere is exactly the type of intractable exploration problem that empowerment maximization wishes to avoid.

We consider a two-stage rollout scheme where we only consider effective empowerment maximization at reachable states. During data collection, we roll out both policies within a single trajectory. The potential policy first moves the agent to a handoff state s_0 , after which the agent switches “on” the skills by sampling a skill z and executing $\pi_{\text{eff}}(\cdot | \cdot, z; s_0)$ (Fig. 1). The two phases serve distinct roles. The potential policy determines *where* the agent initiates skills, while the skill policy measures the controllable futures available from that state. Both of these behaviors are important. The potential policy enables the agent to modulate the starting state for the skills and estimate the empowerment *field*. Meanwhile, skills lower bound the empowerment of any fixed state. We can interpret skills as local probes of empowerment, where the potential empowerment policy ascends the empowerment landscape given by these probes. The two-phase rollout naturally constrains the skill learning to the reachable states and provides a useful training signal for the potential policy in large environments.

4.1 Mutual Information Skill Learning

Contrastive Successor Features. We consider variants of Zheng et al. [12, contrastive successor features (CSF)], a recent state-of-the-art skill-learning method, to lower-bound the effective empowerment $\mathcal{E}_{\text{eff}}(\pi) = I^\pi(Z; S_+ \mid S_0 = s_0)$ for a fixed starting state s_0 . CSF estimates the one-step MI $I(Z; S, S')$ from a contrastively-trained log density ratio; we instead train the classifier on transitions separated by a geometric random variable length of time $\tau \sim \text{Geom}(1 - \gamma)$, which gives the long-horizon MI conditioned on the handoff state. Fixing the skill prior to $p(z \mid s_0) \triangleq p(z) = \text{Unif}(\mathcal{Z})$ removes the otherwise intractable $p(z \mid s_0)$ from the estimator (Definition 3.1). The full derivation is in Appendix A.1.

Following CSF [12] and METRA [11], we parametrize the classifier logits with a bilinear form in terms of representations ϕ :

$$f_\phi(z; s_0, s_+) = z^\top (\phi(s_+) - \phi(s_0)) \approx \log \frac{q_\phi(z \mid s_0, s_+)}{p(z)}. \quad (11)$$

This parameterization forces skills to spread in linearly-separable directions in representation space; the geometric benefits of such a parameterization are discussed in [11]. We train f_ϕ with InfoNCE [22] when $\mathcal{Z}$ is continuous and multi-class cross-entropy when the skill space is finite. Importantly, the classifier may not receive the same set of information as the policy and critic: for instance, we may stipulate that skills are only identifiable if they induce certain changes over certain degrees of freedom, while policies and critics take the entire state into account.

Effective-policy value functions. For a fixed classifier f_ϕ , define the normalized effective empowerment reward

$$r_{\text{eff},\phi}(s, a, s'; z, s_0) \triangleq (1 - \gamma) f_\phi(z; s_0, s'). \quad (12)$$

In other words, the reward measures the relative probability that a transition arose from the executed z (rather than the marginal z). The corresponding value functions are

$$Q_{\text{eff},z}^\pi(s, a; s_0) \triangleq \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k r_{\text{eff},\phi}(s_{t+k}, a_{t+k}, s_{t+k+1}; z, s_0) \mid s_t = s, a_t = a \right], \quad (13)$$

$$V_{\text{eff},z}^\pi(s; s_0) \triangleq \mathbb{E}_{a \sim \pi(a \mid s, z; s_0)} [Q_{\text{eff},z}^\pi(s, a; s_0)]. \quad (14)$$

Training the skills. Averaging the start-state values over skills gives an immediate interpretation of these value functions. The value function is the variational lower bound on the effective empowerment

$$\mathbb{E}_{p(z)} [V_{\text{eff},z}^\pi(s_0; s_0)] \leq \mathcal{E}_{\text{eff}}(\pi, s_0). \quad (15)$$

Under the assumptions of Theorem A.1, alternating policy improvement with on-policy classifier fitting improves the true MI. This variational lower bound has been shown previously in [5].

Training the potential policy. By definition of potential empowerment, the effective empowerment of any policy from any starting state lower bounds the potential empowerment at that same state. Thus, a variational lower bound on effective empowerment lower bounds the potential empowerment:

$$\widehat{\mathcal{E}}_{\text{pot}}(s) \triangleq \mathbb{E}_{p(z)} [V_{\text{eff},z}^\pi(s; s)] \leq \mathcal{E}_{\text{eff}}(\pi_{\text{eff}}, s) \leq \mathcal{E}_{\text{pot}}(s). \quad (16)$$

The potential empowerment serves as the reward for the potential empowerment policy π_{pot} . We train π_{pot} using $\widehat{\mathcal{E}}_{\text{pot}}$ as its intrinsic reward, where $\widehat{Q}_{\text{pot}}^\pi$ and $\widehat{V}_{\text{pot}}^\pi$ are the corresponding value functions. In our experiments, we use SAC for the continuous-control settings [23] and PPO for the discrete settings [?].

In summary, we follow the general formula of using CSF, a scalable MISL method, to lower bound effective empowerment, or the mutual information realized by skill-conditioned policies. Because the effective empowerment, in turn, is a lower bound on potential empowerment, these skills serve as local probes of the potential empowerment. Then, we train a second potential empowerment policy that explicitly ascends this probed landscape, where this agent navigates to states *from which* it can exert control.

4.2 Empowerment in High-Dimensional Settings

With these policy and value function objects in hand, we address two important implementation details that enable the method to scale.

Hindsight relabeling. The effective empowerment policy $\pi_{\text{eff}}(a \mid s; z, s_0)$ conditions on both the skill and the handoff state. This space can be enormous. To reuse previously collected trajectories, we sample an earlier handoff state $\tilde{s}_0$ from the same trajectory and relabel the skill as $\tilde{z} \in \arg \max_z f_\phi(z; \tilde{s}_0, s_+)$. We update the effective policy and critic using the relabeled tuple, following the hindsight-relabeling principle of Andrychowicz et al. [24]. Unfortunately, hindsight relabeling introduces bias, and, in practice, overly concentrates skills, reducing diversity. Thus, we weight each relabeled sample by the inverse effective number of samples of the assigned skill following [?] which helps reduce skill collapse while improving sample efficiency.

Classifier bottleneck. In continuous state spaces, small perturbations can yield high MI without visually distinguishable behavior. We therefore add Gaussian noise to selected state coordinates before passing s_0 and s_+ to the classifier. This is effectively a simplified version of VIB [25] that regularizes the information from the state space to the representation considered in empowerment maximization.

5 Maximizing empowerment in high-dimensional, continuous settings

We evaluate the method in large discrete and continuous environments. We ask:

1. Can we tactably maximize empowerment in large, high-dimensional environments?
2. Does π_{pot} seek states that, intuitively, have high controllability? Does π_{eff} learn distinguishable skills?
3. How does empowerment depend on the chosen representation?

We include training setup and hyperparameters in Appendix B.

Scalability. The method works in high-dimensional, discrete settings. Craftax-Classic [13] is a large and partially observed setting with a 1345-dimensional symbolic observation and 17 discrete actions (Fig. 2a). Using only an empowerment reward defined over the agent’s inventory and a discrete set of 16 skills, the agent unlocks on average 10.6 ± 0.1 out of the 22 total achievements in a single two-phase rollout (mean $\pm$ s.e.m. across 5 seeds, 10^9 environment steps). This intrinsic reward exceeds C-TeC’s reported summed achievement success rate at the same environment-step budget [?]. The potential empowerment estimate also rises steadily over training to 1.9 ± 0.1 nats of the $\ln 16 \approx 2.8$ nat ceiling (Figs. 4g and 4h). Implementation details and hyperparameters for Craftax-Classic are in Appendix C.

To test whether the method works in continuous settings, we chose two high-DoF benchmarks: AntSoccer [27] (Fig. 3b) and BuilderBench [26] (Fig. 2b). Both environments represent common RL benchmarks for locomotion and manipulation and lead to different types of empowering behaviors. In both settings, the empowerment policies learn visually distinguishable and interpretable behavior that spread over the state space (Figs. 2b and 3).

Behavioral interpretation. The AntSoccer environment consists of an ant and a ball. The policy π_{pot} learns to navigate to the *center of the space and stay there* (white, Fig. 3a) while the policies π_{eff} branch and cover the space (colors, Fig. 3a). Importantly, the behavior π_{eff} becomes more distinguishable because the agent first ascends the potential empowerment landscape: the agent is more potentially empowered at the center of the arena, because the agent has learned how to execute distinguishable skills from the center. This result supports our intuition that potential and effective empowerment policies support each other, where effective empowerment policies act as probes of potential empowerment. The average (lower bound on the) estimated potential MI grows over training time as the ant learns to walk, demonstrating that the policy is successfully maximizing the empowerment objectives (Figs. 4a and 4b).

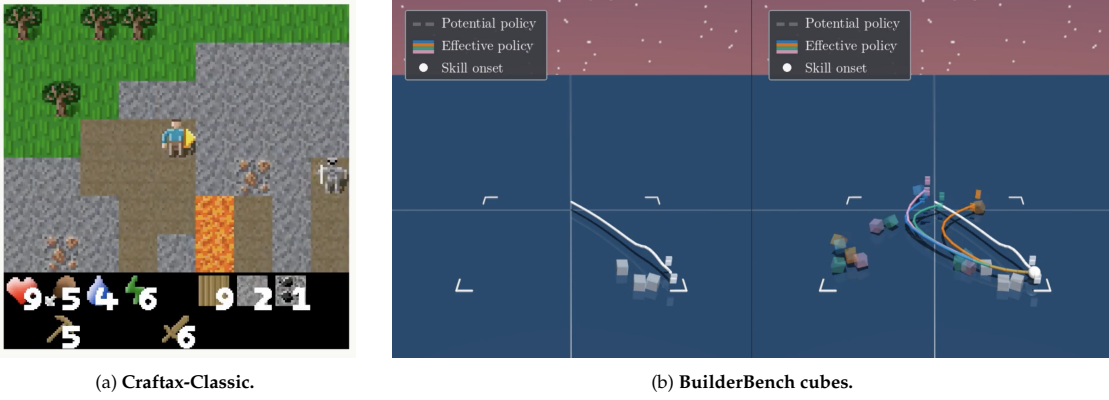


Figure 2: **Empowerment maximization in large discrete and continuous environments.** (a) Potential empowerment policy in Craftax-Classic [13]. The policy learns to gather resources, craft tools, then use tools to collect more resources. (b) Potential and effective empowerment policies in BuilderBench [26] with cubes in the corner. (Left) The potential policy navigates the gripper behind the cubes and stays there, preparing to control the cubes. (Right) The effective policies knock the cubes in different directions.

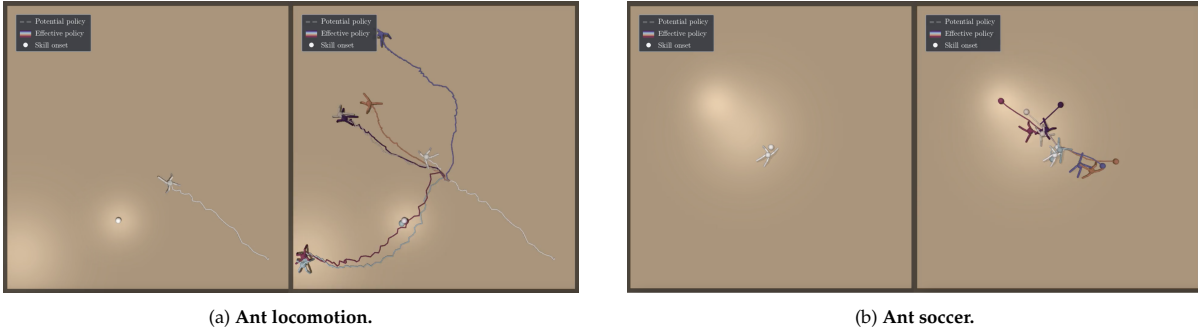


Figure 3: **Empowerment maximization on high-DoF Ant tasks [27].** (a) Ant locomotion: skills cover diverse movement directions, and the potential policy navigates to states with max directional diversity. (b) Ant soccer: skills kick ball in different directions, and the potential policy positions the agent to kick and dribble the ball.

BuilderBench [26] tests manipulation and consists of a gripper with cubes. The environment we consider contains 3 cubes in the corner of a boxed play area (Fig. 2b, white corners denote the boxed play area). Defining the empowerment over the cube xyz positions leads to similar controllability-seeking behavior. The potential empowerment policy sneaks behind the blocks, where it has easy access to the blocks, and stays there. The effective empowerment policies then knock the blocks in different directions. The potential empowerment estimate increases over the course of training as the arm learns to interact with the blocks (Figs. 4e and 4f).

Empowerment depends on the representation. While the above results are encouraging, empowerment can exploit fine-grained perturbations in the chosen representation. This becomes particularly problematic in continuous or large settings – e.g. skills can maximize MI while perturbing the environment in small, meaningless ways. In general, the representations that parameterize empowerment can drastically change which states are central.

For example, we consider an experiment restricting the representation underlying empowerment in AntSoccer. If the encoder must identify skills from changes in the position of the soccer ball, the agent is only empowered if it learns to manipulate and control the ball. Rather than navigating to a physically central state within the arena, the potential empowerment policy now positions the ant to approach and kick the ball, which, after the agent learns to interact with the ball, becomes the new centralized state (Fig. 3b, left; Appendix B.2.2). This behavior is markedly different from the full-observation setting,

where the centralized state is simply the center of the arena, where the agent can easily execute skills such as running in arbitrary directions. Conversely, in Craftax Classic, we consider a similar experiment that *extends* observations from the inventory to the entire observation space, which includes degrees of freedom such as agent orientation and position. While MI quickly approaches $2 \text{ nats} < \ln 16 \text{ nats}$, the agent no longer needs substantial resource accumulation or crafting to achieve high empowerment, since the agent is equally happy with simpler options including running in particular directions, controlling its rate of health depletion, and so on.

These experiments demonstrate that the usefulness of empowerment in large environments depends on the underlying representation. By itself, the empowerment objective does not distinguish between semantically meaningful and incidental controllable degrees of freedom. It is therefore natural to hypothesize that intelligent agents maximize empowerment over degrees of freedom *that align with a form of preference*. Thus, these experimental results suggest that a useful notion of empowerment should reward control over degrees of freedom selected by a separate representation learning process.

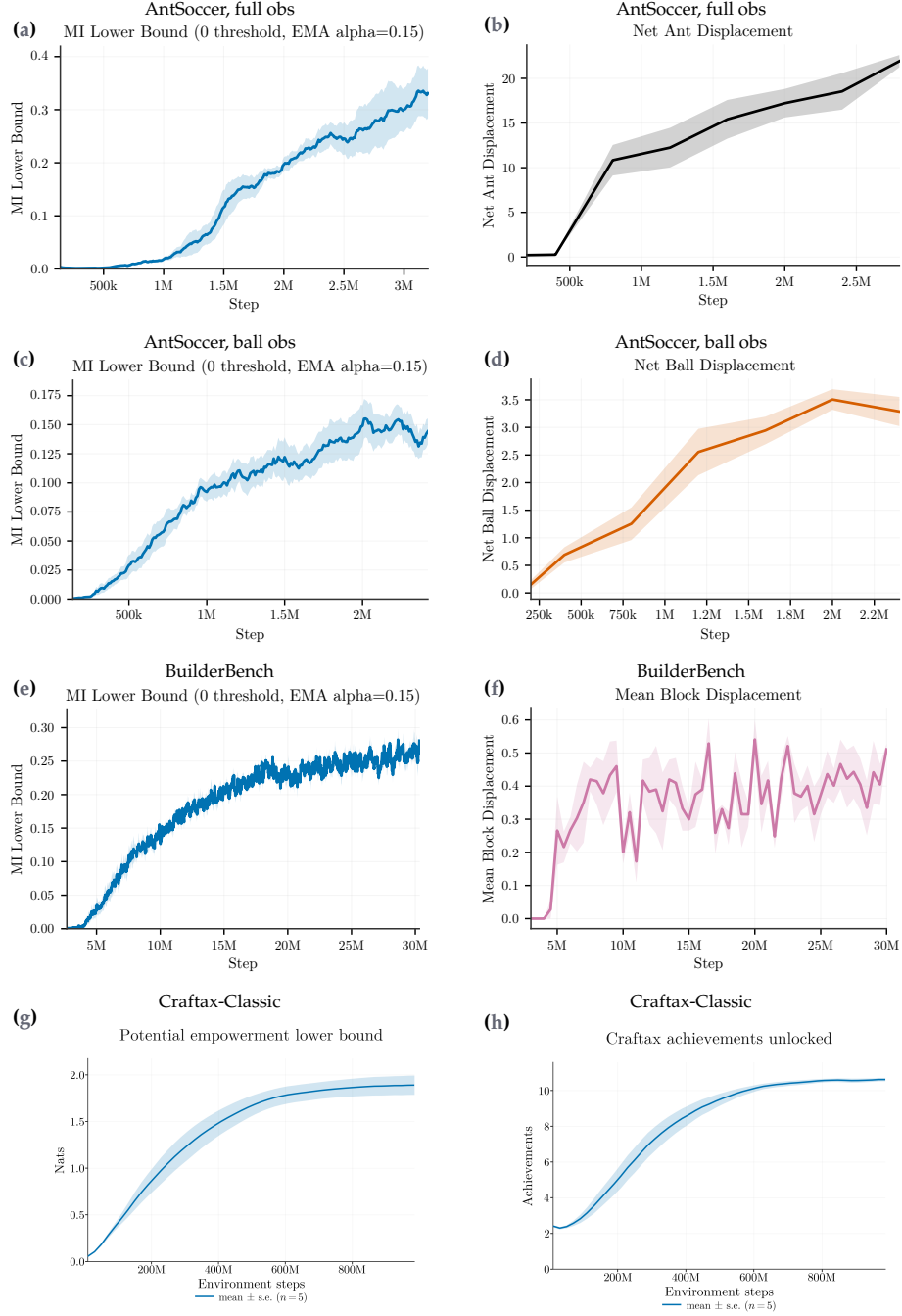


Figure 4: **The average potential empowerment estimate increases with control.** The estimated MI lower bound on potential empowerment increases over environment steps across all settings (left). Behavioral outcomes also improve: displacement of the controlled objects grows in the continuous settings (b, d, f), and the number of achievements unlocked grows in Craftax-Classic (h). The y-axis of the MI panels is in nats. The agent receives no external reward. (a, b): AntSoccer with full observations, over 5 seeds. (c, d): AntSoccer with observations restricted to the ball xy position, over 5 seeds. (e, f): BuilderBench with observations restricted to the cube xyz positions, over 4 seeds. (g, h): Craftax-Classic with the empowerment classifier restricted to the agent’s inventory, over 5 seeds.

6 Discussion

Despite the conceptual appeal of empowerment as a tool for understanding natural agents and building artificial agents, its seeming computational intractability has caused prior work to resort to approximations that do not learn one policy that “keeps its options open” [20].

In providing a conceptual link between skill learning methods and empowerment, we provide a method that maximizes potential empowerment. We present key practical considerations to learn empowerment maximization policies in high-DoF settings. Finally, we present the results of empowerment maximization in a large environment, demonstrating that empowerment rewards control-seeking behavior distinct from that of common intrinsic reward methods. Future work should investigate simplified parameterizations of empowerment maximization that can resolve practical considerations detailed within our method, at scale.

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A Algorithm Details

A.1 Training classifiers to optimize effective empowerment

We build on Zheng et al. [12, CSF], a recent state-of-the-art skill-learning method, to lower-bound the effective empowerment $\mathcal{E}_{\text{eff}}(\pi) = I^\pi(Z; S_+ | S_0 = s_0)$ for a fixed starting state s_0 .

The original CSF objective estimates the one-step mutual information $I(Z; S, S')$, where $s \sim p^\pi(s)$ is the marginal DSOM and s' is the next state. Let $q_\phi(z) = \text{Unif}(\mathcal{Z}) = p(z)$ and fix the starting state. Setting the one-step reward to the log ratio

$$r(s, a, s'; z) = \log \frac{q_\phi(z | s, s')}{p(z)} \quad (17)$$

yields an objective that is a variational lower bound on $I(Z; S, S')$ [12]:

$$\begin{aligned} \mathcal{O}_{\text{csf}}(\pi) &= \mathbb{E}_{p(z)} \sum_t \gamma^t r(s_t, a_t, s_{t+1}; z) \\ &= \mathbb{E}_{p^\pi(s, s', z)} \left[\log \frac{q_\phi(z | s, s')}{p(z)} \right] \\ &\leq I(Z; S, S'). \end{aligned} \quad (18)$$

Following CSF, the log-ratio estimator is trained with standard noise-contrastive objectives (e.g., InfoNCE) [12, 22]. The *conditional, long-horizon* MI, however, would require the different log ratio

$$\log \frac{q_\phi(z | s_0, s_+)}{q_\phi(z | s_0)}, \quad s_+ \sim p_z^\pi(s_+ | s_0), \quad z \sim p(z | s_0).$$

whose denominator $p(z | s_0)$ is generally intractable for high-dimensional z under standard contrastive estimation. This dependence turns out to be unnecessary: for continuous skill spaces, fixing $p(z | s_0) \triangleq p(z) = \text{Unif}(\mathcal{Z})$ and absorbing state-dependent skill selection into the policy loses no generality [28]. The uniform prior therefore removes the intractable $p(z | s_0)$ from the estimator entirely, so the long-horizon conditional MI becomes contrastively estimable. This is *not* generally true for discrete skill spaces, where a uniform prior over K skills restricts the source to the $1/K$ -quantized simplex [28].

The extension to long-horizon effective empowerment is then direct: rather than a classifier predicting z from one-step transitions, we train a classifier predicting z from transitions separated by a geometric random variable length of time $\tau \sim \text{Geom}(1 - \gamma)$. A sample of the one-step reward given (s_0, z, s') approximates

$$r_{\text{eff}}(s, a; z, s_0) \approx \mathbb{E}_{p(s' | s, a)} \left[\log \frac{q_\phi(z | s_0, s_+ = s')}{p(z)} \right], \quad (19)$$

which, in expectation over discounted timesteps and skills, gives $\mathcal{O}(\pi) \leq I^\pi(Z; S_+ | S_0 = s_0) = \mathcal{E}_{\text{eff}}(\pi)$. This is a small modification to CSF (Eq. (18)), yielding a practical method for effective empowerment maximization.

A.2 Policy improvement for the MI that measures empowerment

The result below uses the standard variational information-maximization bound of Barber and Agakov [29], as used for skill discovery in [?], together with classical policy improvement [30]. It shows that alternating policy improvement with classifier fitting is a valid policy-improvement scheme for the *true* MI objective, not merely its variational surrogate:

Theorem A.1 (Policy improvement for the MI that measures empowerment). *Let $0 < \gamma < 1$ be the discount factor and let $\pi_n(a | s, z; s_0)$ be the current skill-conditioned policy at iteration n , with $z \sim p(z)$ sampled at s_0*

and held fixed. Fit an on-policy classifier $q_n(z \mid s_0, s_+)$ to $p^{\pi_n}(z \mid s_0, s_+)$ and define the $(1 - \gamma)$ -normalized action value

$$Q_{\text{eff},z}^{\pi_{\text{eff},n},q_n}(s,a;s_0) \triangleq (1 - \gamma) \mathbb{E}_{\pi_{\text{eff},n}(a|s,z;s_0)} \left[\sum_{k=0}^{\infty} \gamma^k \log \frac{q_n(z \mid s_0, s_{t+k+1})}{p(z)} \middle| s_t = s, a_t = a \right].$$

Let π_{n+1} be obtained by a greedy policy-improvement step with respect to $Q_z^{\pi_n}(s, a; s_0)$ for every z and s_0 . If the classifier is perfectly fit on-policy at each iteration, $q_n(z \mid s_0, s_+) = p^{\pi_n}(z \mid s_0, s_+)$, then

$$I^{\pi_{\text{eff},n+1}}(Z; S_+ \mid S_0) \geq I^{\pi_{\text{eff},n}}(Z; S_+ \mid S_0) \quad \forall n$$

which proves policy improvement under the assumptions of a perfectly-fit classifier.

This is the theoretical license for the practical loop: even though the intrinsic reward is itself policy-dependent (via the classifier), each alternation monotonically improves the true empowerment objective, provided the classifier is refit on-policy. The proof is in Appendix D.1.

In practice, with an estimate of $V_z^{\pi_n}$, improving the *potential empowerment-seeking behavior* reduces to standard RL with intrinsic reward given by the skill-based MI:

$$\widehat{\mathcal{E}}_{\text{pot}}^n(s) \approx \mathbb{E}_{p(z)} \left[V_{\text{eff},z}^{\pi_{\text{eff},n}}(s; s) \right] \approx I^{\pi_{\text{eff},n}}(Z; S_+ \mid S_0 = s) \quad (20)$$

$$\leq \mathcal{E}_{\text{pot}}^n(s) = \max_{\pi \in \Pi_{\mathcal{Z}}} I^{\pi}(Z; S_+ \mid S_0 = s), \quad (21)$$

which, under the γ -discounted sum, lower-bounds the cumulative potential empowerment objective:

$$\mathcal{O}_{\text{p}}^n(\pi_{\text{pot}}) \leq \mathbb{E}_{\pi_{\text{pot}}} \left[\sum_t \gamma^t \mathcal{E}_{\text{pot}}^n(s) \right].$$

The skill-conditioned effective empowerment policy $\pi_{\text{eff}}(a \mid s, z; s_0)$ and the potential policy $\pi_{\text{pot}}(a \mid s)$ are thus closely related but distinct.

A.3 Pseudocode

The discrete and continuous skill settings share the same loop, with f_ϕ parametrized as in Eq. (11), and differ only in how f_ϕ is fit: a multi-class cross-entropy loss over the M skills when $\mathcal{Z}$ is discrete, and a contrastive loss (InfoNCE [22], following CSF [12]) when $\mathcal{Z}$ is continuous. In both cases the policy update uses f_ϕ directly as the reward.

Algorithm 2: MISL for empowerment maximization

Require: MDP $\mathcal{M}$, discount γ , skill space $\mathcal{Z}$, skill prior $p(z) = \text{Unif}(\mathcal{Z})$, rollout length M , iterations N , buffer capacities $|\mathcal{B}_q| \ll |\mathcal{B}_\pi|$

- 1: **Initialize:** $\pi_{\text{eff}}(a | s; z, s_0)$, $\pi_{\text{pot}}(a | s)$, $f_\phi(z; s_0, s_+)$, $\mathcal{B}_q \leftarrow \emptyset$ (small), $\mathcal{B}_\pi \leftarrow \emptyset$ (large)
- 2: **for** $n = 1, \dots, N$ **do**
- 3: **Rollout:** run π_{pot} for M steps, ending at handoff state s_0
- 4: **for** skills $z_k \stackrel{\text{i.i.d.}}{\sim} p(z)$, $k = 1, \dots, K_z$ **do**
- 5: run $\pi_{\text{eff}}(a | s, z_k, s_0)$ from s_0 for αM steps
- 6: sample s_+ from the discounted future states of this skill rollout
- 7: append (s_0, z_k, s_+) to $\mathcal{B}_q$ and $(s_t, a_t, s_{t+1}, s_0, z_k)$ to $\mathcal{B}_\pi$
- 8: **Classifier update:** fit $f_\phi(z; s_0, s_+) \approx \log \frac{p(z|s_0, s_+)}{p(z)}$ on $(s_0, z, s_+) \sim p_{\mathcal{B}_q}$: cross-entropy if $\mathcal{Z}$ discrete, contrastive if $\mathcal{Z}$ continuous
- 9: **Effective-policy update:** sample $s_+ \sim p_{\mathcal{B}_\pi}(s_+)$, HER sample the handoff state s_0 backwards in time from the same trajectory, and assign $z \leftarrow \arg \max_{z' \in \mathcal{Z}} f_\phi(z'; s_0, s_+)$
- 10: update π_{eff} on $Q_{\text{eff}, z}^{\pi_{\text{eff}}}(s, a; s_0)$ with reward $r_{\text{eff}, n} = f_\phi(z; s_0, s_+)$
- 11: **Potential-policy update:** update π_{pot} via SAC [23] on minibatches from $\mathcal{B}_\pi$, training $\hat{Q}_{\text{pot}}^{\pi_{\text{pot}}}$ with intrinsic reward $\widehat{\mathcal{E}}_{\text{pot}}(s) \approx \mathbb{E}_{p(z)}[V_{\text{eff}, z}^{\pi_{\text{eff}}}(s; s)]$
- 12: **return** $\pi_{\text{eff}}, \pi_{\text{pot}}$

B Continuous Experiment Implementation Details

B.1 Practical considerations to scale potential empowerment maximization

Drawing skills from the uniform distribution. As described in Appendix A.1, we roll out skills from a marginal, *state un-conditioned* uniform distribution $z \sim p(z)$ rather than fitting the conditional source $p(z | s_0)$. This contrasts with Variational Intrinsic Control [?], which assumes and fits arbitrary source distributions $p(z | s_0)$. Furthermore, VIC does not simultaneously learn a potential empowerment policy, that is, a policy that maximizes empowerment as defined in [7].

Necessity of Information Bottlenecks information bottleneck (IB). In practice, maximizing the start state-conditioned MI in continuous settings leads to a failure mode: very small, almost imperceptible perturbations in state in, e.g., a deterministic settings can already saturate MI if using a constrained skill space. In fact, high MI *does not directly imply interesting behavior in continuous settings*.

Thus, it is critical to apply a form of information bottleneck from $S_0, S_+ \rightarrow Z$ when computing intrinsic rewards in the continuous settings. Adding Gaussian noise to states before feeding them into the classifier during training. This is a simplified case of applying the Variational Information Bottleneck [25]).

B.2 Hyperparameters and training results

B.2.1 AntSoccer, full obs

Here empowerment is defined over every degree of freedom, the coordinates of both the ant and the ball:

$$\mathcal{E}_{\text{eff}}(\pi) = I^\pi(Z; (S_{+, \text{ant}}, S_{+, \text{ball}}) | (S_{0, \text{ant}}, S_{0, \text{ball}})).$$

In this limit π_{pot} navigates to the center of the space and stays there (white, Fig. 3a), while the skills π_{eff} branch and cover the space (colors). The skills become more distinguishable because the agent first ascends the potential empowerment landscape. The estimated MI lower bound and the net agent displacement both grow over training, despite no external reward for locomotion.

Hyperparameters are in Table 1. Training curves are in Fig. 4a and Fig. 4b.

Table 1: Hyperparameters in AntSoccer, full obs

Parameters	Value
Environment	AntSoccer arena (navigate, single task)
Agent	potential empowerment
State dimension	42
Action dimension	8
Ball observations only	False
Center ball on reset	False
Agent start center fraction	0
Observation noise std (classifier bottleneck)	0.5
Number of skills K	30
Skill dimension d	4
Discrete skills	True
Potential policy steps M	100
Max episode steps	400
Online env steps	5,000,000
Updates per env step	1
Replay buffer size	5,000,000
Skill buffer size	100,000
Value telescoping reference	fixed handoff state
Encoder reference	current state
Discrete reward softmax	True
Achieved-skill relabeling	True (effective-number weighting [?])
Target entropy	0 (learned temperature, init 1)
Learning rate	3e-4
Batch size	256
Critic target update rate	0.02
Discount γ	0.99
Contrastive temperature	0.1
Encoder target update rate	0.99
Separate skill-0 actor and critic	True

B.2.2 AntSoccer, ball obs only

Hyperparameters are in Table 2. Training curves are in Fig. 4c and Fig. 4d.

B.2.3 BuilderBench

Hyperparameters are in Table 3. The figures of MI over environment steps and displacement over environment steps are in Fig. 4e and Fig. 4f.

Table 2: Hyperparameters in AntSoccer, ball obs only

Parameters	Value
Environment	AntSoccer arena (navigate, single task)
Agent	potential empowerment
State dimension	42
Action dimension	8
Ball observations only	True
Center ball on reset	True
Agent start center fraction	0.8
Observation noise std (classifier bottleneck)	0.05
Number of skills K	30
Skill dimension d	4
Discrete skills	True
Potential policy steps M	100
Max episode steps	400
Online env steps	5,000,000
Updates per env step	1
Replay buffer size	5,000,000
Skill buffer size	100,000
Value telescoping reference	fixed handoff state
Encoder reference	current state
Discrete reward softmax	True
Achieved-skill relabeling	True (effective-number weighting [?])
Target entropy	0 (learned temperature, init 1)
Learning rate	3e-4
Batch size	256
Critic target update rate	0.02
Discount γ	0.99
Contrastive temperature	0.1
Encoder target update rate	0.99
Separate skill-0 actor and critic	True

Table 3: Hyperparameters in BuilderBench

Parameters	Value
Environment	BuilderBench (3 cubes, corner cluster)
Agent	potential empowerment
State dimension	51
Action dimension	5
Cube observations only	True
Observation noise std (classifier bottleneck)	0.005 ($\approx 2.5\%$ of cube half-extent)
Number of skills K	30
Skill dimension d	9
Discrete skills	True
Potential policy steps M	100
Max episode steps	400
Online env steps	40,000,000
Env steps per update	8
Parallel environments	100
Replay buffer size	40,000,000
Skill buffer size	2,000,000
Value telescoping reference	fixed handoff state
Encoder reference	current state
Discrete reward softmax	True
Achieved-skill relabeling	True (argmax, effective-number weighting [?])
Target entropy	-12.5 (learned temperature, init 1)
Discount γ	0.99
Contrastive temperature	0.1
Encoder target update rate	0.99
Separate skill-0 actor and critic	True

Table 4: Hyperparameters in Craftax-Classic

Parameters	Value
Environment	Craftax-Classic (symbolic)
Agent	potential empowerment (Potential-PPO)
Observation dimension	1345
Number of discrete actions	17
Parallel environments	8192
Encoder observations	inventory only
Observation noise std (classifier bottleneck)	0
Encoder norm cap c	5.0
Logit temperature τ	0.1 (fixed)
Encoder learning rate	3e-4
Encoder target update rate	0.99
Representation archive size	10,000,000
Encoder updates per policy update	4
Number of skills	16
Skill dimension	16 (standard-basis library)
Discrete skills	True
Potential policy steps	150
Max episode steps	1000
Online env steps	1,000,000,000
Reward	centered logit (??)
Skill-0 reward	normalized potential estimate (Eq. (16))
GAE cut at phase switch	True
Separate skill-0 actor and critic	True
Learning rate	2.5e-4
Batch size	4096
Minibatches	4
Update epochs	4
Discount γ	0.99
GAE λ	0.95
PPO clip ϵ	0.2
Value loss coefficient	0.5
Entropy coefficient	0.01
Max gradient norm	0.5
Actor hidden dims	[512, 512, 512, 512]
Layer norm	True

D Proofs

D.1 Policy improvement and MI lower bound

Theorem A.1 (Policy improvement for the MI that measures empowerment). *Let $0 < \gamma < 1$ be the discount factor and let $\pi_n(a \mid s, z; s_0)$ be the current skill-conditioned policy at iteration n , with $z \sim p(z)$ sampled at s_0 and held fixed. Fit an on-policy classifier $q_n(z \mid s_0, s_+)$ to $p^{\pi_n}(z \mid s_0, s_+)$ and define the $(1 - \gamma)$ -normalized*

action value

$$Q_{\text{eff},z}^{\pi_{\text{eff},n},q^n}(s,a;s_0) \triangleq (1-\gamma) \mathbb{E}_{\pi_{\text{eff},n}(a|s,z;s_0)} \left[\sum_{k=0}^{\infty} \gamma^k \log \frac{q_n(z | s_0, s_{t+k+1})}{p(z)} \middle| s_t = s, a_t = a \right].$$

Let π_{n+1} be obtained by a greedy policy-improvement step with respect to $Q_z^{\pi_n}(s,a;s_0)$ for every z and s_0 . If the classifier is perfectly fit on-policy at each iteration, $q_n(z | s_0, s_+) = p^{\pi_n}(z | s_0, s_+)$, then

$$I^{\pi_{\text{eff},n+1}}(Z; S_+ | S_0) \geq I^{\pi_{\text{eff},n}}(Z; S_+ | S_0) \quad \forall n$$

which proves policy improvement under the assumptions of a perfectly-fit classifier.

The proof separates two facts that are easy to conflate. Holding the classifier fixed gives an ordinary discounted MDP, so policy improvement applies. Refitting the classifier makes the variational MI bound tight again.

Proposition D.1 (A fixed classifier defines an ordinary intrinsic-reward MDP). *Fix a classifier $q(z | s_0, s_+)$ and define the normalized intrinsic reward*

$$r_{\text{eff},q}(s_0, z, s') \triangleq (1-\gamma) \log \frac{q(z | s_0, s')}{p(z)}.$$

For any policy π , define

$$Q_z^{\pi,q}(s,a;s_0) \triangleq \mathbb{E}_{\pi(a|s,z;s_0)} \left[\sum_{k \geq 0} \gamma^k r_q(s_0, z, s_{t+k+1}) \middle| s_t = s, a_t = a \right].$$

Then $Q_{\text{eff},z}^{\pi,q}$ satisfies the Bellman equation

$$\begin{aligned} Q_{\text{eff},z}^{\pi,q}(s,a;s_0) &= (1-\gamma) \mathbb{E}_{p(s'|s,a)} \left[\log \frac{q(z | s_0, s')}{p(z)} \right] \\ &\quad + \gamma \mathbb{E}_{p(s'|s,a)} \left[V_{\text{eff},z}^{\pi,q}(s'; s_0) \right], \end{aligned}$$

where $V_{\text{eff},z}^{\pi,q}(s; s_0) = \mathbb{E}_{\pi(a|s,z;s_0)} Q_{\text{eff},z}^{\pi,q}(s,a;s_0)$.

Proof. Re-express the discounted sum in terms of the reward and remaining summation. The classifier is fixed, so $r_{\text{eff},q}$ is a stationary reward for the induced MDP indexed by (z, s_0) . $\square$

Proposition D.2 (Barber–Agakov lower bound is tight at the on-policy posterior). *For any policy π and any classifier $q(z | s_0, s_+)$,*

$$\begin{aligned} \mathcal{L}(\pi, q) &\triangleq \mathbb{E}_{p(z) p_z^\pi(s_+ | s_0)} \left[\log \frac{q(z | s_0, s_+)}{p(z)} \right] \\ &\leq I^\pi(Z; S_+ | S_0 = s_0). \end{aligned}$$

Equality holds if $q(z | s_0, s_+) = p^\pi(z | s_0, s_+)$.

Proof. Write the MI as

$$I^\pi(Z; S_+ | S_0 = s_0) = H(Z) + \mathbb{E}_{p^\pi(s_+ | s_0)} \mathbb{E}_{p^\pi(z | s_0, s_+)} [\log p^\pi(z | s_0, s_+)].$$

For every s_+ ,

$$D_{\text{KL}}(p^\pi(z | s_0, s_+) \| q(z | s_0, s_+)) \geq 0,$$

so replacing $\log p^\pi$ by $\log q$ gives a lower bound. Since $H(Z) = -\mathbb{E}_{p(z)} \log p(z)$ and the joint law factors as $p(z) p_z^\pi(s_+ | s_0)$, the bound is exactly $\mathcal{L}(\pi, q)$. Equality holds at the on-policy posterior. $\square$

Proof of Theorem A.1. At iteration n , the classifier q_n is fit to the on-policy posterior, so

$$\mathcal{L}(\pi_{\text{eff } n}, q_n) = I^{\pi_{\text{eff } n}}(Z; S_+ | S_0).$$

Now hold q_n fixed. By Proposition D.1, this is an ordinary MDP with stationary intrinsic reward. A greedy policy-improvement step therefore gives

$$\mathcal{L}(\pi_{\text{eff } n+1}, q_n) \geq \mathcal{L}(\pi_{\text{eff } n}, q_n).$$

Finally refit the classifier to the new on-policy posterior. By Proposition D.2, the refitted classifier makes the lower bound tight and can only increase the lower-bound value relative to the old classifier:

$$\begin{aligned} I^{\pi_{\text{eff } n+1}}(Z; S_+ | S_0) &= \mathcal{L}(\pi_{\text{eff } n+1}, q_{n+1}) \\ &\geq \mathcal{L}(\pi_{\text{eff } n+1}, q_n) \\ &\geq \mathcal{L}(\pi_{\text{eff } n}, q_n) \\ &= I^{\pi_{\text{eff } n}}(Z; S_+ | S_0). \end{aligned}$$

Thus exact on-policy classifier fitting plus greedy improvement monotonically improves the true MI. $\square$

Glossary

$\mathcal{E}_{\text{eff}}$ the effective empowerment as a function of the policy . 4, 6, 13, 15

$\mathcal{E}_{\text{pot}}$ The potential empowerment at state s and iteration n , defined as the maximal possible mutual information between skills and future states under some policy. . 14

$\mathcal{E}_{\text{pot}}$ the potential empowerment as a function of the state . 3–6, 14, 15, 22

π_{eff} an effective empowerment maximizing policy . 4–7, 14, 15, 21, 22

π_{pot} a potential empowerment maximizing policy . 4–7, 14, 15

Acronyms

CSF contrastive successor features. 6, 13, 14

DSOM discounted state occupancy measure. 3, 13

IB information bottleneck. 15

MDP Markov decision process. 2, 3, 15

MI mutual information. 2–4, 6, 7, 13–15, 20

MISL mutual information skill learning. 2–5, 15

RL reinforcement learning. 2, 14